Year 8
Mathematics

Unit 3: Working with Powers

3.3 Factorising Expressions

Factorising Expressions

STARTER · 5 MIN

Date: _____ **Title: Factorising Expressions**

LEARNING OBJECTIVES

- To be able to factorise an algebraic expression

STARTER *Answer in your book. No calculators.*

1. Expand $2(x + 3)$
2. Expand $9(2 - x)$
3. Expand $4(3x + 2)$
4. Expand $5(4 - 7x)$

Starter — Answers

STARTER

Tick or correct in green pen

1. $2x + 6$
2. $18 - 9x$
3. $12x + 8$
4. $20 - 35x$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can identify the common factor in each term	☐
SILVER	I can write the expression as a product using brackets	☐
GOLD	I can check my answer by expanding the brackets	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Factorising Expressions

INTRODUCTION · 3 MIN

Factorising reverses expanding: take the HCF outside a bracket.

Common factor

Divides every term exactly.

$6x$ and 4: common factor 2

Step 1

Find the HCF of the terms.

$6x + 4 \rightarrow$ HCF 2

Step 2

HCF outside, divide each term inside.

$6x + 4 = 2(3x + 2)$

Check

Expand your answer.

$2 \times 3x + 2 \times 2 = 6x + 4$

Examples 1 and 2 — Numbers as the Common Factor

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Find the HCF Check by expanding

Example 1

Factorise

$6x + 4$

Example 2

Factorise

$12 + 18x$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

HCF of 6 and 4 is 2

$6x + 4 = 2(3x + 2)$

Check $= 6x + 4$

Tip: $6x \div 2 = 3x$, $4 \div 2 = 2$.

Example 2

HCF of 12 and 18 is 6

$12 + 18x = 6(2 + 3x)$

Check $= 12 + 18x$

Tip: $2(6 + 9x)$ is not fully factorised.

Task 1

PRACTICE · 9 MIN

BRONZE *Factorise*

Hint Find the HCF Check by expanding

1. $4x + 8$

2. $14m + 28$

3. $20 - 10s$

4. $8y + 12$

5. $12n - 6$

★ $4a + 10b$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $4(x + 2)$

2 $14(m + 2)$

3 $10(2 - s)$

4 $4(2y + 3)$

5 $6(2n - 1)$

★ $2(2a + 5b)$

Examples 3 and 4 — Letters in the Common Factor

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Letters can be common factors Lowest power of x

Example 3

Factorise

$5x^2 + 15x$

Example 4

Factorise

$pq + 2p$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

HCF of $5x^2$ and $15x$ is $5x$

$$5x^2 + 15x = 5x(x + 3)$$

Tip: $5x^2 \div 5x = x$.

Example 4

Both terms contain p

$$pq + 2p = p(q + 2)$$

Tip: check by expanding.

Task 2

PRACTICE · 9 MIN

SILVER *Factorise completely*

Hint Letters can be common factors Lowest power of x

1. $9x^2 + 12x$

2. $x^2 + 7x$

3. $10x^2 - 15x$

4. $3y^2 + y$

5. $2m + n + 8 + m$

★ $8ab - 12a$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $3x(3x + 4)$

2 $x(x + 7)$

3 $5x(2x - 3)$

4 $y(3y + 1)$

5 $3m + n + 8$ (no common factor)

★ $4a(2b - 3)$

Examples 5 and 6 — Factorising Completely

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Completely: use the HCF Area ÷ width = length

Example 5 (2 marks)

Factorise completely

$4x^3 + 12x^2$

Example 6 (2 marks)

$12x + 24b$ can be factorised in 5 ways. Find them all.

Which one is factorised completely?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

HCF: 4 and x^2

$$4x^3 + 12x^2 = 4x^2(x + 3)$$

Tip: take the lowest power of x.

Example 6

Use each common factor

$2(6x + 12b)$, $3(4x + 8b)$

$4(3x + 6b)$, $6(2x + 4b)$

Completely $= 12(x + 2b)$

Tip: completely means the HCF.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: factorise completely*

Hint Completely: use the HCF Area ÷ width = length

1 $15x^3 - 5x$

2 $32x + 16x^2$

3 $3x^3 + 6x$

4 $12y^4 - 4y^3$

5 Ali writes $6x + 12 = 2(3x + 6)$.
Is it fully factorised?

★ Area $8x + 20$.
Width 4. Find the length.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $5x(3x^2 - 1)$

2 $16x(2 + x)$

3 $3x(x^2 + 2)$

4 $4y^3(3y - 1)$

5 No: $6(x + 2)$

★ $2x + 5$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Factorise $6x + 9$ and $8x + 12$

SILVER Factorise $4a + 10b$ and $12x - 6$

GOLD Factorise $5x^2 + 15x$ and $9x^2 + 12x$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **$3(2x + 3)$ and $4(2x + 3)$**

SILVER **$2(2a + 5b)$ and $6(2x - 1)$**

GOLD **$5x(x + 3)$ and $3x(3x + 4)$**

Exam Practice — Factorising

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Find the HCF Check by expanding

Question 1

(2 marks)

Factorise completely $12 + 18x$

.....

Question 2

(2 marks)

Factorise

(a) $5y + 20$ (1)

.....

(b) $14 - 21a$ (1)

.....

Exam Practice — Letters as Factors

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Letters can be common factors
Lowest power of x

Question 3

(2 marks)

Factorise completely $10x^2 - 15x$

.....

Question 4

(2 marks)

Factorise completely $4x^3 + 12x^2$

.....

Exam Practice — Reasoning

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Completely: use the HCF
Area ÷ width = length

Question 5

(2 marks)

Sam says $8x + 16 = 2(4x + 8)$ is factorised completely.

Is he right? Explain.

.....

Question 6

(3 marks)

A rectangle has area $6x^2 + 9x$ and width $3x$.

(a) Find its length. (2)

.....

(b) Find the length when $x = 4$. (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

6(2 + 3*x*) B2

B1 for 2(6 + 9*x*) or 3(4 + 6*x*)

Tip: the HCF is 6.

Question 2

(a) 5(*y* + 4) B1

(b) 7(2 − 3*a*) B1

Tip: check by expanding.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$5x(2x - 3)$ B2

B1 for $5(2x^2 - 3x)$ or $x(10x - 15)$

Tip: HCF $5x$.

Question 4

$4x^2(x + 3)$ B2

B1 for a partly factorised answer

Tip: HCF $4x^2$.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

No: $4x + 8$ still has factor 4 — **B1**

$8(x + 2)$ — **B1**

Tip: completely means use the HCF.

Question 6

(a) $6x^2 + 9x = 3x(2x + 3)$ — **M1**

Length $2x + 3$ — **A1**

(b) 11 — **B1**

Tip: area = width × length.